Lunar tidal potential measured with a coastal–inland clock comparison

Optical clocks provide a direct probe of gravitational potential through the relativistic frequency shift. Following its temporal variation requires separating a weak geophysical signal from changes in the clocks and their comparison. Here we analyse a Shanghai–Wuhan strontium–ytterbium comparison over a 1,350-km fibre link, sampled at different lunar–solar phases between September 2025 and August 2026. Eight observation periods contain 546.4 accumulated sample-hours. A joint model with one offset per period gives a semidiurnal response of 1.02 ± 0.09 relative to the combined solid-Earth and ocean-loading prediction. Separating the nearby bands yields an M2+N2 lunar response of 0.99 ± 0.12 and an S2+K2 response of 1.13 ± 0.29. These are model-conditional statistical uncertainties; original-group and segment-offset alternatives give similar lunar scales. Phase-diverse sampling reduces the response to coherent solar-day disturbances, while explicit disturbance gains and separate daily-controlled analyses delimit that interpretation. The coastal–inland geometry gives a substantial predicted ocean-loading contribution and extends contemporaneous clock-network studies to an intermittently observed baseline across the annual cycle. The results support clock comparisons as probes of changing tidal potential and identify the calibration and sampling conditions needed for more general geophysical measurements.

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Optical clocks translate a difference in gravitational potential into a measurable frequency ratio.[1,2,3] Comparisons have progressed from laboratory redshift tests to centimetre-scale geodesy, transportable clocks and intercity networks.[4,5,6,7] Time-dependent potential adds a further observable: the changing gravitational environment of the clocks, complementary to acceleration and displacement measurements.[8,9,10] Tides provide a structured prediction against which both the phase and scale of this response can be tested.

A coastal–inland baseline is informative because solid-Earth deformation and ocean loading contribute differently at its endpoints. Temporal coverage is equally consequential. The dominant lunar semidiurnal constituent M2 advances through solar time, changing its relative phase with nearby S2. Observations distributed over that evolution help distinguish their template amplitudes. They also provide a way to quantify, rather than assume away, leakage from solar-day disturbances.

We use the Shanghai Sr–Wuhan Yb comparison reported in the related frequency-ratio experiment by Hou and colleagues.[11] Its clock systems and optical dissemination build on separately documented clock evaluations and fibre-transfer techniques.[12,13,14,15,16] Those apparatus references establish context; performance measured on another link is not assigned to the present observations. According to

the experimental team, the observations were acquired independently of the contemporaneous European network experiment of Stahl and colleagues and before its public preprint.[17] The present synthesis was prepared with knowledge of that work. Here we test the physical scale of a coastal–inland tidal response and separate its lunar semidiurnal component using phase-diverse observations.

A phase-diverse coastal–inland record

The retained record comprises 54 effective segments in 29 original operating groups and eight observation periods: September and October 2025, January, April, May and late June–July 2026, followed by two August periods. Its 1,966,908 retained counter samples correspond to 546.4 sample-hours at the nominal one-second gate. The 3,299 accepted ten-minute bins contain 545.9 sample-hours. These exposure measures describe intermittent observations, not uninterrupted stability or eight independent experimental replications (Fig. 1).

The physical prediction combines solid-Earth and ocean-loading contributions.[18,19,20,21,22] For M2, their differential amplitudes are 3.43×10^{-18} and 1.42×10^{-18}, with a relative phase of 19.8°. Their vector sum is 4.78×10^{-18}. These are predicted components; the measurement tests their combined response. Fig. 1 places the observation periods alongside the astronomical S2–M2 phase, showing how the intermittent record samples different spring–neap configurations.

The primary model uses one constant offset for each observation period, denoted P8. Group-resolved clock corrections enter the summer ratio reconstruction. Earlier measurements are represented on period-relative scales because their absolute frequency-chain reconstruction is incomplete. A free period offset absorbs a period-constant shift; it does not automatically absorb changes between groups within that period. Alternative group and segment offsets test sensitivity to this choice without claiming to validate every missing correction.

A common semidiurnal response

We jointly fit the bins with a semidiurnal template and a separate diurnal template. The semidiurnal coefficient retains the predicted relative proportions of M2, N2, S2 and K2. No free exact 24-h or 12-h sinusoid is included in this primary model. The result is

$$A_{\mathrm{SD}} = 1.02 \pm 0.09, \qquad A_{\mathrm{D}} = 0.67 \pm 0.36.$$

Errors are the larger of residual-scaled least-squares and leave-one-UTC-day-out jackknife errors.[23] They are conditional on the specified tidal synthesis, weighting and clock corrections, whose uncertainties are not propagated. The reduced chi-square is 1.15. Fig. 2 shows an illustrative summer group after removal of its fitted period offset, alongside the unscaled physical template; all bins enter the joint

inference.

The semidiurnal scale changes little when the offset resolution is increased: original-group offsets give 1.01 ± 0.10, and separate segment offsets give 0.98 ± 0.10 (Extended Data Table 1). A hybrid model with early group offsets and summer period offsets gives 1.01 ± 0.10. These are comparisons of the same measurements. Their consistency supports the period-level representation for this estimator, rather than establishing absolute calibration between periods.

Separating the lunar response

Splitting the semidiurnal prediction into M2+N2 and S2+K2 yields

$$A_{\mathrm{L}} = 0.99 \pm 0.12, \quad A_{\mathrm{S}} = 1.13 \pm 0.29, \quad A_{\mathrm{D}} = 0.68 \pm 0.36.$$

The first coefficient describes the lunar semidiurnal template. The second describes the near-12-hour template, including the lunisolar K2 constituent; it is not an exclusively solar gravitational measurement. Their least-squares coefficient correlation is -0.36 (Fig. 3). Across the same-data S54, G29, P8 and hybrid offset models, the lunar scale ranges from 0.98 to 1.00. The scale remains conditional on the adopted physical synthesis: the original stored-template alternative gives $A_{\mathrm{L}} = 1.26 \pm 0.18$ and $A_{\mathrm{SD}} = 1.20 \pm 0.14$. That comparison changes the loading synthesis convention and is not attributed solely to the remaining diurnal weighting limitation (Methods).

Phase diversity reduces coherent solar-day leakage, but does not exclude unmeasured disturbances. In the primary design, a sinusoidal fractional-frequency disturbance of amplitude 10^{-18} at exactly 24 h or 12 h can change A_{L} by at most 0.0138 or 0.0142 when its phase is fixed across periods. Allowing independent phases in each period raises these bounds to 0.0330 and 0.1656, respectively. These are estimator gains, not measured environmental amplitudes. The larger second bound motivates the explicit daily-controlled comparisons below.

The semidiurnal feature decreases after tidal regression, whereas a small residual diurnal excess remains relative to the stated white-noise reference (Fig. 3c). The spectrum and component estimates are two views of the same observations, not independent significance tests. The ellipse displays the fitted correlation scaled by reported marginal errors; it is not a calibrated joint confidence region.

Complementary controls and interpretation

All eight period-specific semidiurnal estimates are positive, with unequal uncertainties (Fig. 4). The September 2025 and late-August 2026 estimates are weakly constrained; a common amplitude does not imply equal-strength seasonal detections. On the same summer subset, one common offset gives

0.77 ± 0.16, compared with 0.80 ± 0.16 for separate period offsets. This is a conditional consistency check on the corrected summer levels. It is not a long-term stability measurement.

A separately processed campaign analysis allows independent exact 24-h and 12-h terms and obtains $A_{\mathrm{L}} = 0.86 \pm 0.18$. Splitting summer daily terms by month gives 0.82 ± 0.21 and permitting linear evolution gives 0.85 ± 0.20 (Extended Data Fig. 1). Its selection, overlapping filter and physical template differ from P8. A further 2026-only reconstruction gives 0.89 ± 0.28 with daily terms; its six-period sign test gives $P = 0.0625$ and a broad 95% acceptance interval. These analyses show how the inference changes under explicit daily controls, but their simulation coverage or probabilities cannot be transferred to the primary 0.99 ± 0.12 estimate (Extended Data Table 2).

In the 2026 reconstruction, adding the simultaneously sampled B2 monitor changes the whole-template coefficient by 0.0057 ± 0.0073 on identical support without daily terms. This quantifies the scale change when the recorded monitor is included (Extended Data Fig. 3). It does not exclude disturbances in the clocks, unmonitored short transfer branches or other hardware. No same-site null comparison is available. The accompanying noise simulations and retained-level long-period bounds provide additional, explicitly conditional checks in Methods and Extended Data.

The coastal–inland comparison recovers a semidiurnal response consistent with the predicted physical amplitude and constrains its lunar component with limited dependence on observing-offset resolution. Its contribution complements the European network's combined-tide admittance and the overlapping summer ratio analysis of Hou and colleagues.[17,11] It is neither a statistically independent repetition of that summer dataset nor the same estimator as the European combined-tide result. Phase-diverse sampling, a substantial predicted loading contribution and explicit separation of level calibration from within-period dynamics define the present opportunity. Extending it to separately measured loading or slow geophysical changes will require calibrated levels and stronger constraints on time-varying clock and transfer errors.

References

1. Ludlow, A. D., Boyd, M. M., Ye, J., Peik, E. & Schmidt, P. O. Optical atomic clocks. *Reviews of Modern Physics* **87**, 637–701 (2015). Source
2. Chou, C. W., Hume, D. B., Rosenband, T. & Wineland, D. J. Optical Clocks and Relativity. *Science* **329**, 1630–1633 (2010). Source
3. Mehlstäubler, T. E., Grosche, G., Lisdat, C., Schmidt, P. O. & Denker, H. Atomic clocks for geodesy. *Reports on Progress in Physics* **81**, 064401 (2018). Source
4. Bothwell, T. et al. Resolving the gravitational redshift across a millimetre-scale atomic sample. *Nature* **602**, 420–424 (2022). Source
5. McGrew, W. F. et al. Atomic clock performance enabling geodesy below the centimetre level. *Nature* **564**, 87–90 (2018). Source
6. Grotti, J. et al. Geodesy and metrology with a transportable optical clock. *Nature Physics* **14**, 437–441 (2018). Source
7. Lisdat, C. et al. A clock network for geodesy and fundamental science. *Nature Communications* **7**, 12443 (2016). Source
8. Voigt, C., Denker, H. & Timmen, L. Time-variable gravity potential components for optical clock comparisons and the definition of international time scales. *Metrologia* **53**, 1365–1383 (2016).

Source
9. Denker, H. et al. Geodetic methods to determine the relativistic redshift at the level of 10^{-18} in the context of international timescales: a review and practical results. *Journal of Geodesy* **92**, 487–516 (2018). Source
10. Schröder, S., Stellmer, S. & Kusche, J. Potential and scientific requirements of optical clock networks for validating satellite-derived time-variable gravity data. *Geophysical Journal International* **226**, 764–779 (2021). Source
11. Hou, L. et al. Yb/Sr optical clock comparison at 2.2×10^{-18} uncertainty over a 1350-km fiber link. *arXiv*, 2610.34250v1 (2026). Source
12. Jia, Z.-P. et al. Improved systematic evaluation of a strontium optical clock with uncertainty below 1×10^{-18}. *Metrologia* **63**, 025002 (2026). Source
13. Zhu, Q. et al. Ytterbium lattice clock with systematic uncertainty of 1.3×10^{-18} and instability at the 10^{-19} level. *arXiv*, 2606.10514v2 (2026). Source
14. Riehle, F. Optical clock networks. *Nature Photonics* **11**, 25–31 (2017). Source
15. Chen, F.-X. et al. 10^{-21}-Level optical frequency dissemination over 2067 km of noise-loaded field-deployed fiber network. *Light: Science & Applications* **15**, 276 (2026). Source
16. Schioppo, M. et al. Comparing ultrastable lasers at 7×10^{-17} fractional frequency instability through a 2220 km optical fibre network. *Nature Communications* **13**, 212 (2022). Source
17. Stahl, K. et al. Optical clock network tracks tidal variations of the gravity potential. *arXiv*, 2610.39635v1 (2026). Source
18. Petit, G. & Luzum, B. (eds) IERS Conventions (2010). *IERS Technical Note* **36** (2010). Source
19. LEGOS, NOVELTIS & CLS FES2022 (Finite Element Solution) Ocean Tide. *CNES data set* (2024). Source
20. Lyard, F. H., Allain, D. J., Cancet, M., Carrère, L. & Picot, N. FES2014 global ocean tide atlas: design and performance. *Ocean Science* **17**, 615–649 (2021). Source
21. Farrell, W. E. Deformation of the Earth by surface loads. *Reviews of Geophysics* **10**, 761–797 (1972). Source
22. Han, D. & Wahr, J. The viscoelastic relaxation of a realistically stratified earth, and a further analysis of postglacial rebound. *Geophysical Journal International* **120**, 287–311 (1995). Source
23. Efron, B. & Stein, C. The Jackknife Estimate of Variance. *The Annals of Statistics* **9**, 586–596 (1981). Source

Figure legends

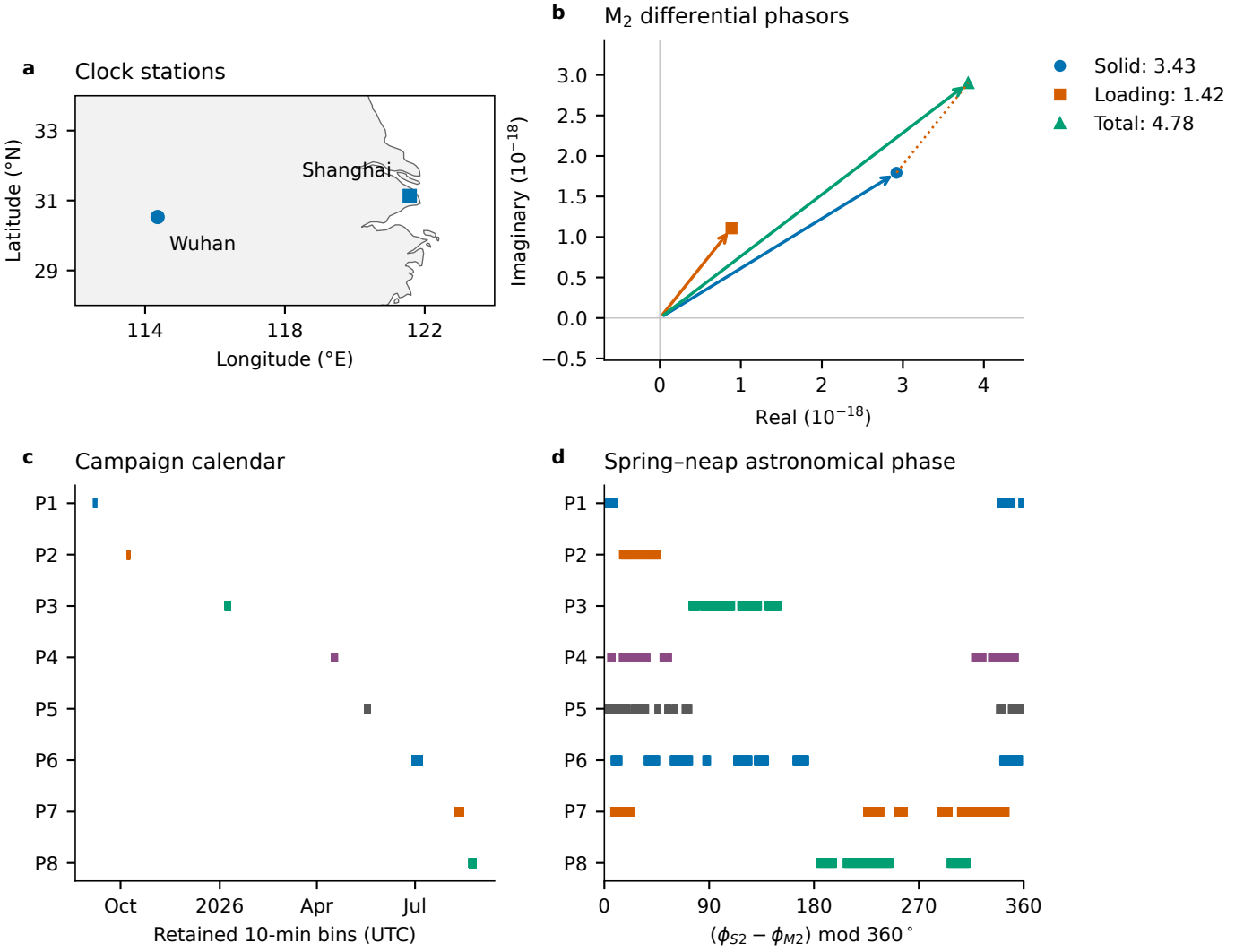


Figure 1: Coastal–inland geometry and phase coverage

Fig. 1 | A coastal-inland clock baseline sampled at different tidal phases. **a**, Clock sites on a local Natural Earth basemap; geographical separation is distinct from the 1,350-km fibre route. **b**, Predicted M2 differential solid-Earth, loading and total phasors in the adopted lag convention. Magnitudes are 3.43, 1.42 and 4.78 in 10^{-18} units. **c**, Locations of the retained bins in eight observation periods. **d**, Astronomical S2–M2 phase, including nodal phase, at the same bins. This phase coverage describes sampling geometry, not an additional fitted observation. The 3,299 bins contain 545.9 sample-hours; 546.4 sample-hours precede the minimum-bin-membership requirement.

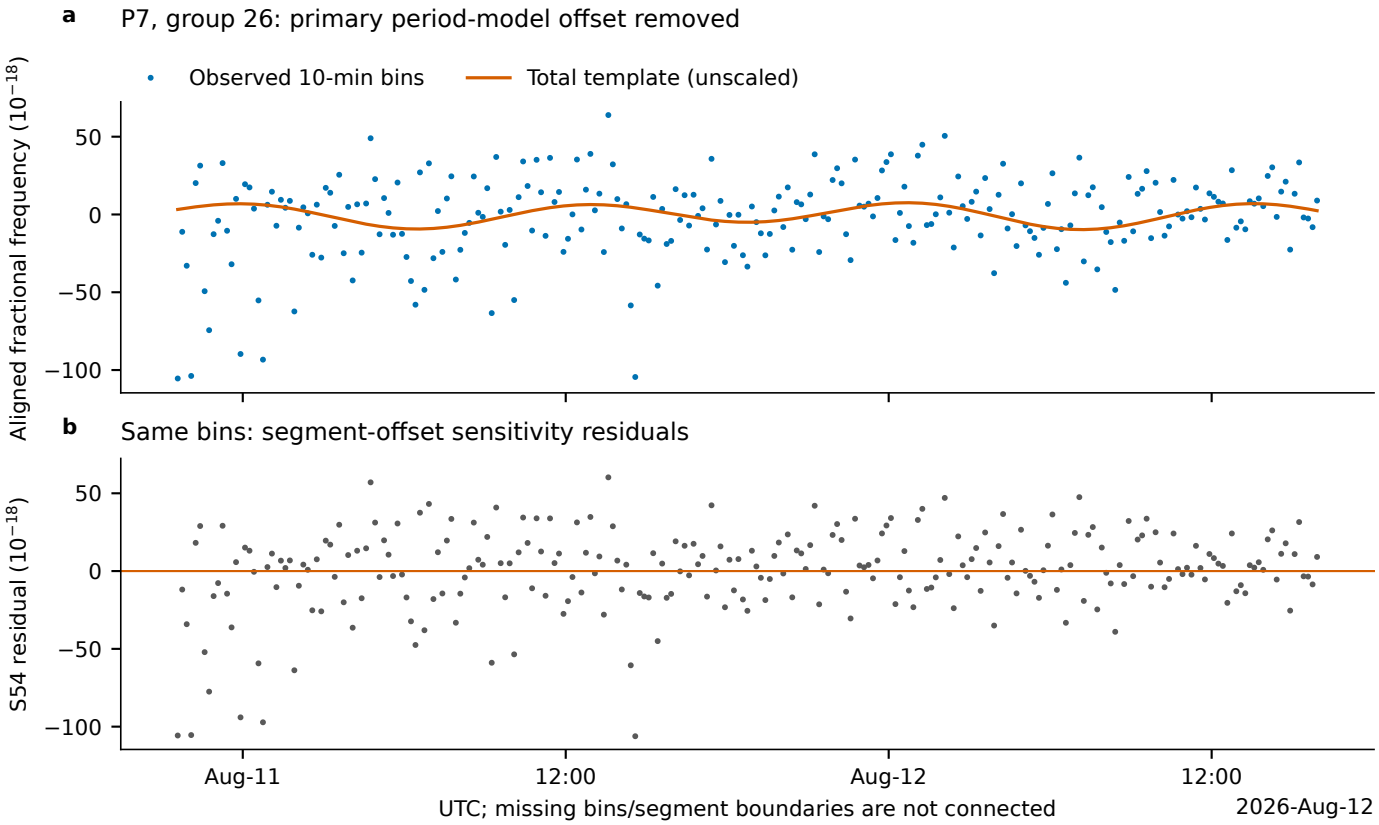


Figure 2: Illustrative summer observations and sensitivity residuals

Fig. 2 | A summer observing group and its tidal prediction. **a**, All 255 retained ten-minute bins from original group 26 in period P7, selected by the longest UTC span among groups in P7 and P8. The primary period-model offset for P7 is removed for display. The total physical template is unscaled. **b**, The same observations under the S54 two-band segment-offset sensitivity fit. Panel b is not panel a minus its orange curve and is not the primary-model residual; it shows the consequence of the alternate offset resolution. Both panels share fractional-frequency limits, without extra display averaging, point removal or connections across missing bins/segment boundaries. This illustrative view does not establish long-term stability; inference uses all 3,299 bins.

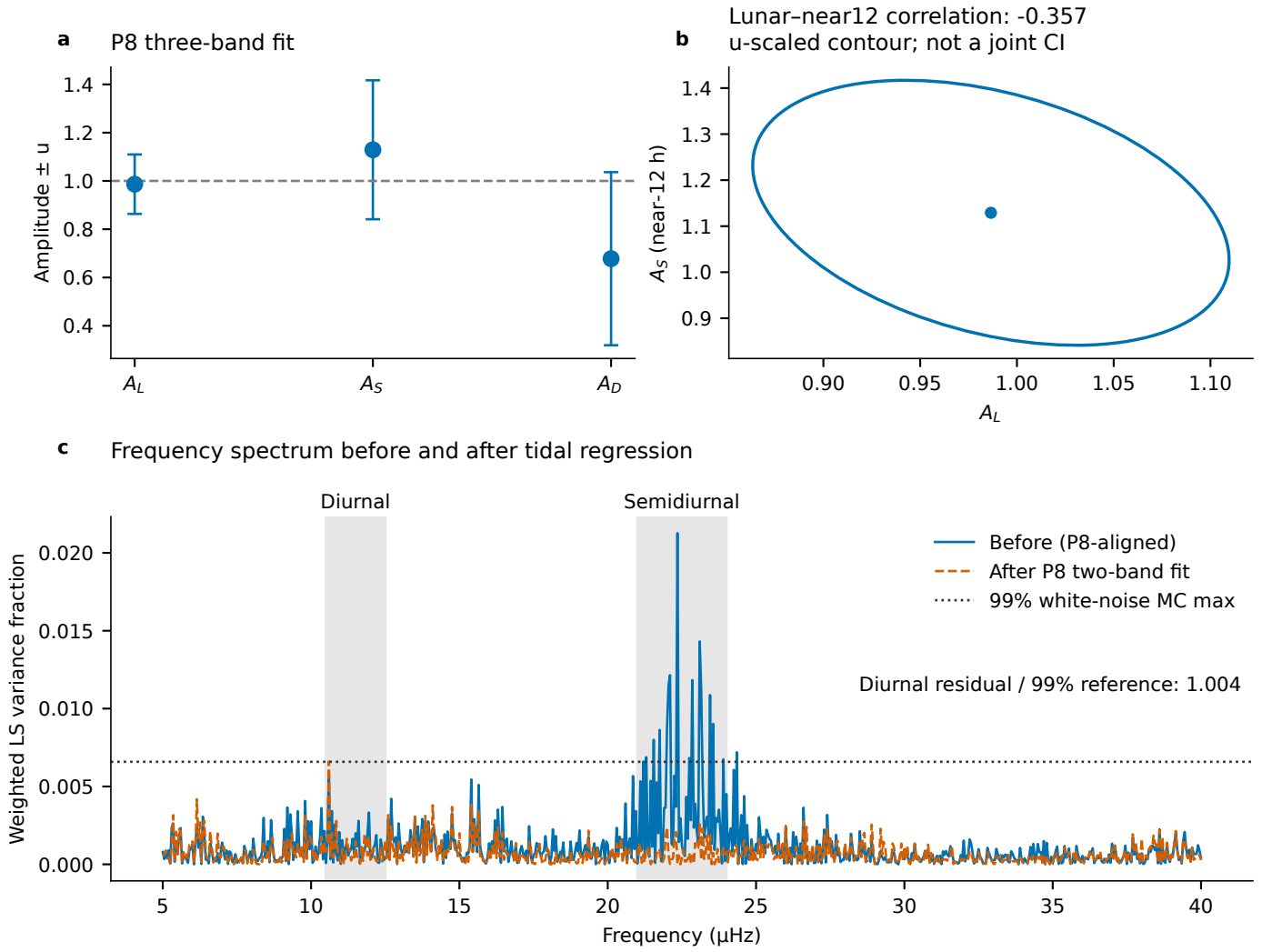


Figure 3: Joint lunar response and spectra

Fig. 3 | Joint lunar and near-12-hour response. **a**, Primary three-band coefficients for M2+N2 (A_L), S2+K2 (A_S, including lunisolar K2) and K1+O1+P1+Q1 (A_D). Error bars are the larger scaled-LS/UTC-day-jackknife errors; the line is unit response. A_D is a model-dependent diagnostic. **b**, Lunar–near-12-hour correlation (−0.36). The contour uses least-squares correlation and reported marginal errors at unit Mahalanobis radius, not a calibrated joint confidence level. **c**, Weighted floating-mean spectral variance fractions before and after the primary two-band regression, with the 99th percentile of maximum power under the specified white-noise simulations. The residual diurnal maximum slightly exceeds that reference. All panels use the same data and are not independent significance tests; neither spectrum is a physical power spectral density.

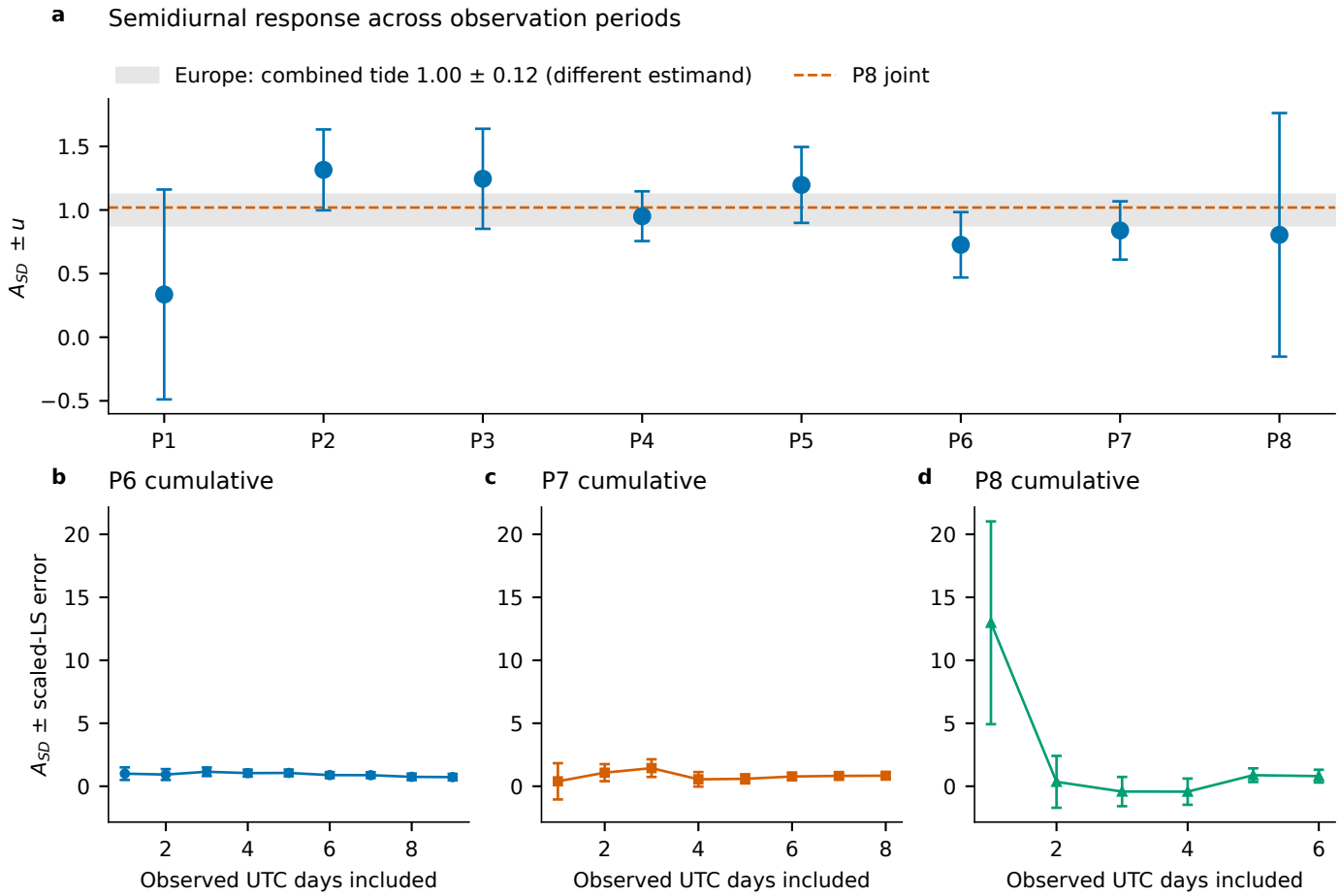


Figure 4: Period estimates and accumulated fits

Fig. 4 | Semidiurnal response across observation periods. **a**, Two-band estimates for P1–P8 with formal max-rule statistical errors and the joint estimate. The shaded European combined-tide admittance 1.00 ± 0.12 is a different estimand, quoted with its reported uncertainty for context only.[17] No precision ranking or statistical pooling follows from interval overlap. **b–d**, Cumulative estimates within P6, P7 and P8 as observed UTC days are added. These descriptive curves use scaled-LS errors only, share observations and retain the early uncertain estimates. They do not establish a general detection time, long-term stability or repeated independent detections. Their endpoint errors may differ from formal max-rule period errors.

Methods

Samples, time labels and averaging

The primary record follows the updated experimental table of 54 effective segments, grouped into 29 operating groups and eight periods. Original group 22 (segment 47) is excluded by that table; segment 48 remains included. Files are concatenated within periods before applying the intervals, so midnight file boundaries do not truncate intervals. Printed Beijing-time labels are UTC+8 h. Selection floors counter labels to the printed second; subsecond jitter may place multiple counter rows in one labelled second. Samples are not fabricated or deduplicated solely by that label.

After removing points farther than 10 Hz from their segment median, 1,966,908 samples remain, compared with 1,966,905 in the experimental table. The three-row difference is recorded in the boundary-second reconciliation; the prescribed total tolerance is ten rows. A shared boundary second belongs to the preceding segment. This is the primary mask; the six-scale local-excursion screen of the separate campaign analysis is not applied here.

Bins are formed within each segment on a fixed ten-minute Beijing-time grid, requiring at least 180 retained samples. Adjacent segments can have separate partial bins with the same centre time without sharing members. The 3,299 bin rows contain 545.924 sample-hours; their row count multiplied by ten minutes is 549.833 h and is not an exposure estimate. Theoretical series are interpolated to bin-centre UTC times without extrapolation. This differs from the auxiliary reconstruction's exact source-member averaging. Periods and segments describe one paired physical comparison, not independent biological or laboratory replicates.

Frequency coordinate and clock corrections

For the early periods, the working fractional-frequency coordinate is

$$y_{\mathrm{rel}} = -\frac{d_p(f_{\mathrm{beat}} - f_{\mathrm{ref},p})}{193399200000000\ \mathrm{Hz}},$$

where $d_p = 2$ in September and October 2025 and $d_p = 1$ in 2026; $f_{\mathrm{ref},p}$ is the retained-beat median for that period. Calculations use units of 10^{-18}. The fitted offsets do not establish an absolute connection across the year. For P1–P5, absent frequency-chain information prevents a complete absolute reconstruction; tabulated shifts are recorded rather than assigned unsupported conversions. The 37.2-mHz feedback entry in P1–P3 is absorbed by the period offset only if it is constant during that period. Early group-dependent Yb terms are not all constant: for example, P3 sums span −9.84 to −9.56 in 10^{-18} units. Their varying part is not absorbed by one period intercept. G29/H comparisons assess sensitivity to group levels, not the completeness of this correction ledger.

For summer P6–P8, the legacy frequency-chain formula[11] is evaluated at 80-digit Decimal precision for

each bin mean, with group-resolved Sr and Yb shifts, then referenced to the mean summer ratio before conversion to floating point. The returned ratio is $R = f_{\mathrm{Yb,WUH}}/f_{\mathrm{Sr,SHA}}$; the legacy function internally forms Sr/Yb and then inverts it. Positive $y = R/\bar{R} - 1$ corresponds to Wuhan Yb becoming faster relative to Shanghai Sr. With positive effective gravitational potential W, its tidal clock contribution is $-[\delta W_{\mathrm{WUH}} - \delta W_{\mathrm{SHA}}]/c^2$. The frozen within-segment sign check gives slope 0.99999858 and signed correlation 0.999999999986, with sign factor +1: no sign reversal was applied to this dataset. This is an algebraic/implementation comparison, not an independent hardware polarity calibration.

The finest available group correction records take precedence over period averages. Summer Sr density, AC Stark, second-order Zeeman and background shifts and Yb corrections enter at that resolution; the Sr background term is -0.67×10^{-18} and black-body radiation is corrected online. Early AC entries are missing. The provisional May Yb convention is inherited from April; its constant level would be absorbed by the May offset, but any unrecorded time dependence would not. Correction uncertainties are not propagated. A possible historical Yb correction near 1.1×10^{-17} is not added a second time without identifying its relationship to the tabulated shifts (Supplementary Methods S1).

Physical template and specified synthesis

The solid-Earth input is the archived IERS-based series; ocean loading uses FES2022 constituents and elastic load responses.[18,19,21,22] The primary synthesis reconstructs loading from the supplied complex load-potential coefficients with pyTMD 3.0.9 astronomical arguments and nodal corrections.[24] Constituent k contributes $f_k\,\mathrm{Re}[C_k \exp(-i\phi_k)]$, where f_k is the nodal factor and ϕ_k includes astronomical and nodal phase. The primary convention was chosen from source-phase checks, not by minimizing distance of the fitted amplitude from one.

The stored alternative loading series used the conjugate phase convention and omitted the nodal factors used here. Its M2 total amplitude is approximately 3.90×10^{-18}, versus 4.78×10^{-18} for the adopted synthesis. The unchanged solid-Earth diurnal weighting also remains incompletely validated: its K1 amplitude is approximately 18% below the potential-weighted comparison. Accordingly, the diurnal coefficient is a model-dependent diagnostic. The whole-template sensitivity reported beside the main results changes multiple template terms and is not assigned solely to K1. Full physical-template certification remains pending.

The fitted bands are M2+N2, S2+K2, and K1+O1+P1+Q1. K2 is lunisolar. Harmonic projection includes additional nuisance constituents to limit leakage. The difference between the total predicted series and the three bands is retained as a fixed residual r_i. Within each band, constituent proportions and phases remain model-specified. Thus the analysis estimates template scales, not unrestricted individual constituent amplitudes or Love numbers.

Estimator, weights and offset alternatives

For bin i, the three-band model is

$$y_i = A_{\mathrm{L}} L_i + A_{\mathrm{S}} S_i + A_{\mathrm{D}} D_i + r_i + c_{p(i)} + \epsilon_i.$$

The two-band model constrains $A_{\mathrm{L}} = A_{\mathrm{S}} = A_{\mathrm{SD}}$. Joint weighted least squares estimates amplitudes and offsets together. P8 uses eight period indicators; S54 uses segment indicators; G29 uses original-group indicators; H uses early group and summer period indicators. A3 uses only summer P6–P8 with one common offset. A3 is compared with a separate-period fit on that same subset, not with a full-record estimate as if only the offsets had changed.

Per-period weights use a short-time white-frequency-noise coefficient α_p estimated from overlapping Allan deviations at 256, 512, 1,024, 2,048 and 4,096 sample intervals.[25] Each eligible one-second segment is processed separately after subtracting the fixed unit-amplitude model and its own mean; segment deviations are length-weighted, and a fixed $\tau^{-1/2}$ slope is used. No fitted tidal amplitude or fitted offset is read, and no Allan window crosses segment boundaries. Constant subtraction cancels algebraically in within-segment differences. The implementation uses row-index sampling at the nominal one-second gate; exact continuity across rejected or missing rows is not established by this calculation. These values define conditional short-time weights, $\sigma_i = \alpha_{p(i)}/\sqrt{n_i}$, and provide no long-term stability result. No stability curve based on fit-adjusted offsets, recentered runs or spliced residuals is retained anywhere in this manuscript or supplement.

Scaled least-squares errors use the residual reduced chi-square. UTC-day deletion removes each observed day and empty offset columns; the larger marginal LS/day error is reported. Period deletion is a separate diagnostic: for A_{L} it gives 0.1314, compared with LS 0.1231 and day deletion 0.1116. The reported max-rule error is not a calibrated joint confidence region. The Fig. 3 contour uses $UR_{\mathrm{LS}}U$, with U the diagonal matrix of reported errors, at unit Mahalanobis radius. Cumulative curves use scaled-LS errors only and share observations.

Solar-day gains and spectral diagnostics

Let X contain the tidal bands and offsets, $W = \mathrm{diag}(\sigma_i^{-2})$, and h_j^{T} be row j of $(X^{\mathsf{T}}WX)^{-1}X^{\mathsf{T}}W$. At exact solar period P, a unit sine/cosine pair Q_P gives fixed-phase gain $G_{j,P} = \|h_j^{\mathsf{T}} Q_P\|_2$. For arbitrary independent phases in periods p, the equal-amplitude bound is $G^{\mathrm{ind}}_{j,P} = \sum_p \|h_j^{\mathsf{T}} I_p Q_P\|_2$. A disturbance of at most $a \times 10^{-18}$ in each period therefore satisfies $|\Delta A_j| \le a G^{\mathrm{ind}}_{j,P}$, conditional on that sinusoidal disturbance class. Unequal amplitudes require the corresponding amplitude-weighted sum. Extended Data Table 1 reports all gains. They do not supply measured disturbance amplitudes or bounds on arbitrary waveforms. Omitting free daily terms reduces parameter count while strengthening assumptions

about unmodelled variation.

The weighted floating-mean Lomb–Scargle diagnostic[26] uses actual bin times and a grid of 701 frequencies from 5 to 40 µHz. The reference is the 99th percentile of full-grid maxima in 200 independent white-noise realizations with bin-specific uncertainties, seed 0. The residual diurnal maximum is 0.00660977, against 0.00658372. This small excess is reported, not used to retune the fit. The ordinate is normalized variance reduction, not a physical power spectral density or a coloured-noise false-alarm probability. Display alignment and same-sample subtraction are not independent confirmation.

Analysis rules were recorded before this branch's first measurement fit; they were not publicly preregistered and do not make the historical study prospective. P8/G29 consistency gates passed; the specified residual spectral excess was non-stopping. Software success does not certify the physical template or confer author approval.

Separate daily-controlled and noise analyses

The campaign analysis associated with the supplied v19 draft uses a distinct retained mask, approximately 544 h, 54 segments, five campaigns and overlapping 1,200-s triangular windows at 600-s stride. Its full physical-potential construction and local excursion screen differ from P8. With segment offsets and campaign-specific exact 24-h/12-h terms, $A_{\mathrm{L}} = 0.86 \pm 0.18$; its no-daily whole-template coefficient is 0.90 ± 0.12. Its 29-group resampling interval is [0.63,1.17]. Extended Data Table 3 preserves all eight solar-day injection rows. The simulation models and tail calculation are specified in Supplementary Methods S5. Their coverage belongs to that estimator, not P8. Source eight-period wild-test values are withheld pending verification of statistic, enumeration and two-sided tie handling.

The v42 auxiliary reconstruction uses 2,905 non-overlapping means from 1,737,073 retained 2026 members, 49 intervals and six periods. It uses member-matched predictions and fixed five-/ten-day baseline indicators, with or without free daily terms. Its ten-day whole-template estimate without daily terms is 0.88 ± 0.10; its daily-controlled lunar estimate is 0.89 ± 0.28. Its uncertainty rule takes the maximum of LS, day and period deletion errors, unlike P8. Restricted wild tests[27] enumerate 64 period-sign patterns under independent symmetric period errors; the minimum two-sided probability is 0.03125. Whole-template and lunar 95% acceptance intervals are [0.51,1.42] and [−1.18,4.22], respectively. Joint lunar phase tests give $P = 0.25$ at zero and 0.72 at unit in-phase response for ten-day baselines. Extended Data Fig. 2 retains the phase display.

Extended Data Tables 4–6 retain block support and all conditional bias/error and coverage rows: 1,024 realizations per Gaussian scenario and 4,096 per Student/MA scenario. White noise, OU correlations at 1 h, 12 h and 3 days, period constants, and heavy-tailed innovations with MA(1) structure probe those specific designs. Seeds and covariance definitions are in Supplementary Methods S3. These checks do not guarantee coverage for arbitrary real noise or for P8.

Monitored transfer and long-period bounds

B2 matching in the 2026 auxiliary reconstruction requires one valid monitor value for every clock-bin member, without interpolation. Excursion limits of ±10, ±100 and ±1,000 Hz around the recorded 10-MHz carrier retain 1,476, 1,519 and 1,520 complete bins. Identical-support paired day deletions give the change after adding B2: 0.0057 ± 0.0073 without daily terms and 0.0415 ± 0.0345 with daily terms at the reference screen. The monitor's own template projections are 0.074 ± 0.087 and 0.54 ± 0.46. Normalization uses 193399200000000 Hz without an assumed one-way factor of two. Extended Data Fig. 3a shows nominal 95% intervals with 19 degrees of freedom. This is a control on a specified monitored channel; unmonitored clock and laboratory-transfer effects remain outside it, and ±5-s matching sensitivity does not certify synchronization accuracy.

The long-period auxiliary fits preserve reconstructed inter-period levels and use one global intercept, joint Mf/Mm/Msf/Sa/Ssa quadratures, short-period tides and global daily terms. They do not adjust individual segment or month centres. Conditional bounds and the distinct global-drift sensitivity are specified in Supplementary Methods S4. The nearly equal station latitudes suppress the predicted seasonal differential signal to about 2×10^{-20}, far below the conditional bounds. No seasonal detection or measured long-term stability is claimed.

Availability and assistance

The primary manuscript source package contains derived bins, templates, constituent coefficients, correction records, masks, reconciliation and figure sources. The fitting pipeline resides in the separately versioned period-offset research branch, with an external legacy estimator and frequency-chain module; figure regeneration is not raw-data reproduction. The frozen evidence records Python 3.12.3, NumPy 2.5.3, pandas 2.3.3, SciPy 1.18.1 and pyTMD 3.0.9. The auxiliary workflow has separate source identities, code, seeds and validation records. No primary measurement fit was rerun for this editorial synthesis.

Data availability: author-approved access to raw observations, derived series, correction records and source membership remains to be specified before submission. Code availability: the authors must provide the review-access route and archived versions for all three analysis routes and their external dependencies; no public repository deposit is claimed. The campaign analysis still requires its complete source-member and code release.

OpenAI Codex agents assisted source comparison, static code review, figure-label revision and manuscript assembly. Internal AI-assisted reviews do not constitute independent human approval. Authors are responsible for checking scientific content and completing the journal's disclosure, authorship and access statements.

Methods references

24. Sutterley, T. C. et al. pyTMD: Python-based tidal prediction software, version 3.0.9 (version used in the primary analysis). Source
25. Riley, W. J. Handbook of frequency stability analysis. *NIST Special Publication* **1065** (2008). Source
26. Zechmeister, M. & Kürster, M. The generalised Lomb-Scargle periodogram: a new formalism for the floating-mean and Keplerian periodograms. *Astronomy & Astrophysics* **496**, 577–584 (2009). Source
27. Cameron, A. C., Gelbach, J. B. & Miller, D. L. Bootstrap-Based Improvements for Inference with Clustered Errors. *Review of Economics and Statistics* **90**, 414–427 (2008). Source
28. Folkner, W. M., Williams, J. G., Boggs, D. H., Park, R. S. & Kuchynka, P. The planetary and lunar ephemerides DE430 and DE431. *Interplanetary Network Progress Report* **42-196**, 1–81 (2014). Source

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[Corresponding author and contact details to be supplied.] Supplementary Methods accompany this author-review draft. The combined review PDF includes the Article, Methods, Extended Data and Supplementary Methods; the latter are also supplied as a separable file.

Extended Data

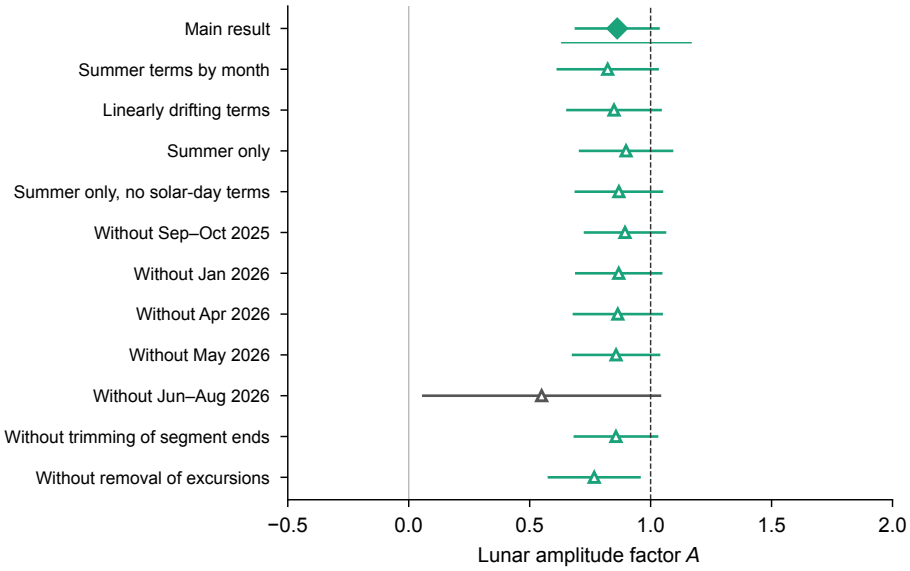


Figure 5: Separate campaign daily-control sensitivity

Extended Data Fig. 1 | Daily-controlled campaign analysis, distinct from P8. Author-supplied campaign sensitivity graphic from the v19 analysis, preserving all plotted values. The reference is the M2+N2 response 0.86 ± 0.18 with segment offsets and independent campaign 24-h/12-h terms. Month-split summer terms, evolving daily terms, summer-only support and campaign deletions test that analysis. Its mask, overlapping triangular averages, template and nuisance design differ from the primary P8 route. Thick bars show the reported working standard errors; the thin bar beneath the reference is the 95% operating-group resampling interval. The final row retains the unscreened result 0.77 ± 0.19. No value or coverage claim is transferred to P8; the correlated fits are not independent.

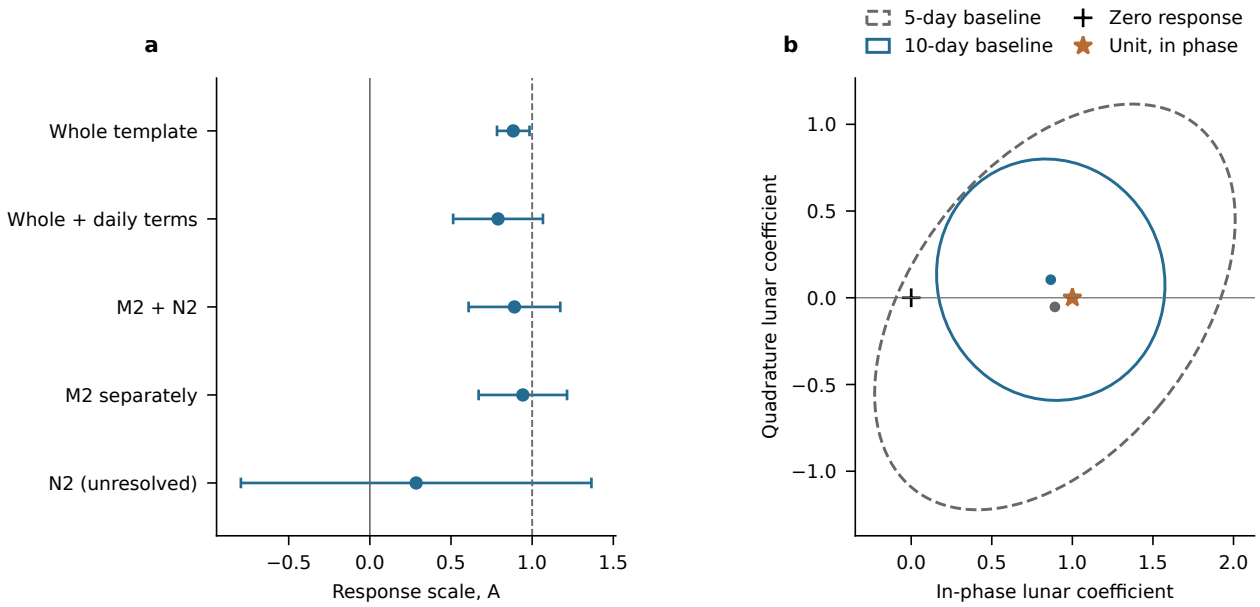


Figure 6: Auxiliary lunar component and phase

Extended Data Fig. 2 | Lunar component and phase in the 2026 reconstruction. **a**, Frozen component estimates under the auxiliary models; errors are their reported working standard errors. **b**, M2+N2 in-phase/quadrature estimates with nominal 95% delete-day ellipses. The unit in-phase point lies inside both displayed ellipses; the origin is inside the five-day ellipse and outside the ten-day ellipse. This is not robust exclusion of zero across baseline choices. The ten-day restricted six-period joint test gives $P = 0.25$ at zero and 0.72 at unit in-phase response. These 2,905-bin auxiliary results use a different estimator and covariance construction from primary Fig. 3b; the contours and probabilities cannot be interchanged.

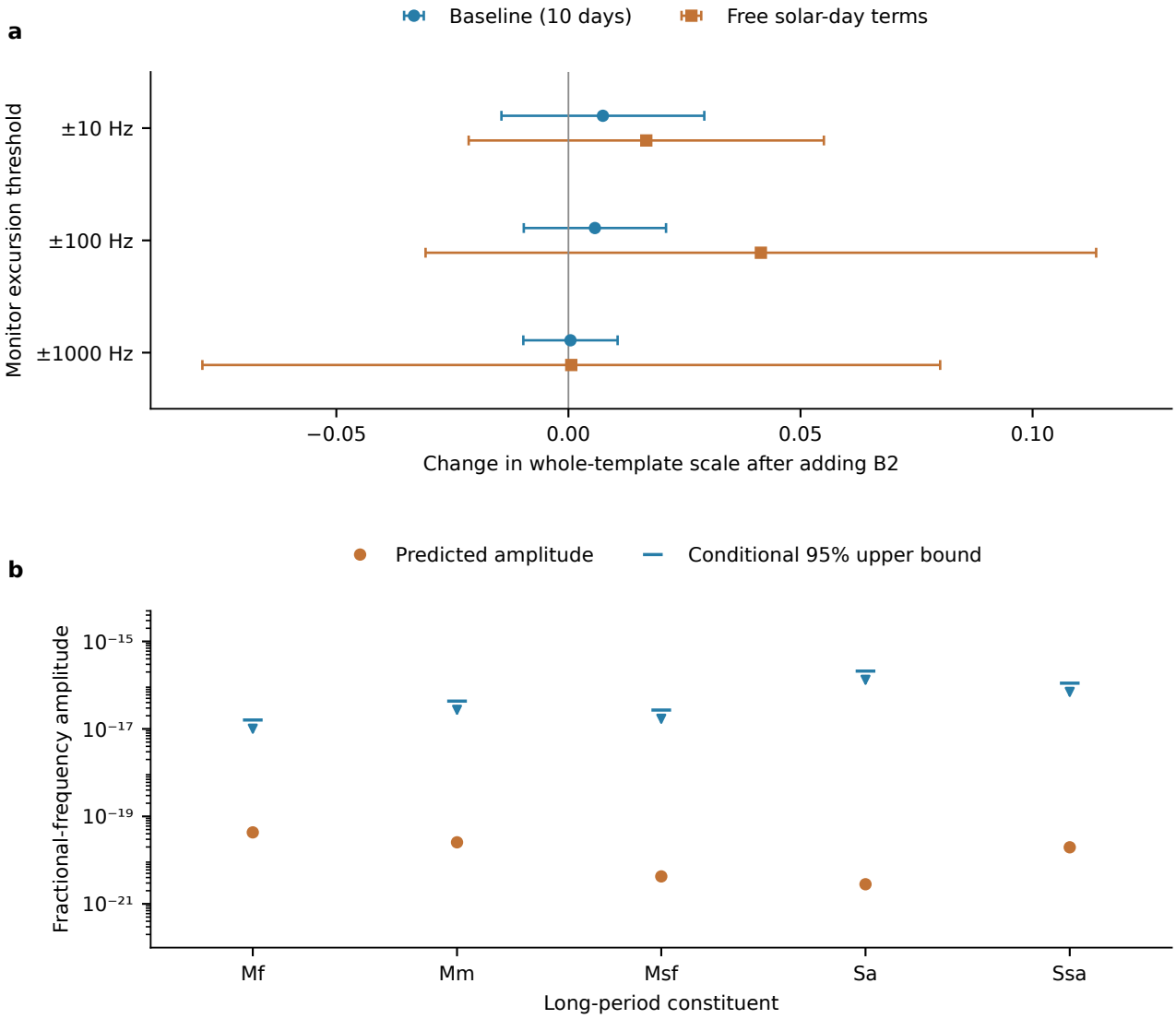


Figure 7: Paired B2 regression and conditional long-period bounds

Extended Data Fig. 3 | Auxiliary monitored-channel and retained-level checks. a, Change in whole-template scale after adding B2 on identical matched support at each monitor threshold. Error bars are nominal 95% paired delete-day intervals with 19 degrees of freedom; all six estimates are retained. The monitor tests only its sampled channel. **b**, Individual nominal 95% upper bounds on Mf, Mm, Msf, Sa and Ssa amplitudes for the 2026 retained-level model, alongside predicted amplitudes. Bounds use a global intercept without segment/month shifts and are conditional on the incomplete calibration and loading ledger. Arrows mark upper bounds; their lengths are graphical. This is an amplitude-bound comparison, not a long-term stability measurement or a seasonal detection.

Extended Data Table 1 | Offset schemes and solar-day gains in the primary dataset. Errors are max(scaled LS, UTC-day jackknife). S54, G29, P8 and H use the same 3,299 bins. A3 and the summer-period comparison use only P6–P8; their difference cannot be interpreted as full-record offset sensitivity. The lower table gives coefficient changes per unit disturbance amplitude of 10^{-18}: fixed phase across periods versus independently chosen period phases, with the same amplitude bound in each period. These are analytic design gains, not measured bias limits.

Scheme / support	A_{SD}	Two-band A_D	A_L	A_S
S54	0.98 ± 0.10	0.58 ± 0.53	1.00 ± 0.13	0.93 ± 0.36
G29	1.01 ± 0.10	0.75 ± 0.42	0.98 ± 0.12	1.09 ± 0.30
P8	1.02 ± 0.09	0.67 ± 0.36	0.99 ± 0.12	1.13 ± 0.29
H	1.01 ± 0.10	0.72 ± 0.37	0.99 ± 0.12	1.08 ± 0.29
A3	0.77 ± 0.16	-0.09 ± 0.42	—	—
Summer, period offsets	0.80 ± 0.16	-0.06 ± 0.43	—	—

Model / coefficient	Disturbance period (h)	Fixed-phase gain	Independent-phase bound
two / SD	24	0.01131	0.02688
two / D	24	0.31133	0.46972
two / SD	12	0.10383	0.13976
two / D	12	0.05678	0.09853
three / L	24	0.01384	0.03296
three / S	24	0.03278	0.09524
three / D	24	0.30958	0.47087
three / L	12	0.01416	0.16558
three / S	12	0.44318	0.47115
three / D	12	0.05151	0.09881

Extended Data Table 2 | Analysis identities and limits of transfer. Each result belongs to its own retained mask, coordinate, prediction and nuisance model. "Campaign" is the v19 processing route; "auxiliary" is the v42 2026 reconstruction. Their tests do not calibrate the P8 error bars. Sample-hours denote accumulated nominal-gate samples, not continuous stability.

Property	Primary P8	Campaign analysis	Auxiliary 2026
Scope	2025–2026; 8 periods	2025–2026; 5 campaigns	2026; 6 periods
Segments / intervals	54	54	49
Retained exposure	546.4 h before bin cut	about 544 h	20.105 sample days
Averaging	3,299 non-overlapping 600-s means	1,200-s triangular filter, 600-s stride	2,905 non-overlapping 600-s means
Prediction sampling	Bin centre	Filtered source prediction	Exact retained members
Primary nuisance	Period offsets; no daily terms	Segment offsets; campaign daily terms	10-day blocks; optional daily terms
Lunar result	0.99 ± 0.12	0.86 ± 0.18	0.89 ± 0.28 with daily terms
Other central target	Semidiurnal 1.02 ± 0.09	Whole tide 0.90 ± 0.12, no daily terms	Whole tide 0.88 ± 0.10, no daily terms
Main error rule	max(LS, day)	max(LS, day) for daily-controlled lunar fit	max(LS, day, period)
Conditional simulations	No route-specific coverage supplied	Solar-day injections (Table 3)	Noise/coverage (Tables 5–6)
Resampling inference	Day/period errors recorded	Group bootstrap [0.63,1.17]	Lunar six-period $P = 0.0625$

Primary short-time weighting coefficients α_p (units $10^{-18}\sqrt{\text{s}}$) are 970.3, 570.4, 429.8, 459.6, 492.8, 507.4, 576.0 and 529.7 for P1–P8. They describe the stated nominal-sample, model-subtracted weighting procedure, not independent long-term clock or link stability.

Extended Data Table 3 | Separate campaign solar-day injection tests. All eight rows reproduce the campaign pipeline's residual-based simulations, with 300 realizations each and seed 20261116. Scenarios (a)–(d) are defined in Supplementary Methods S5. Bias and SD concern the recovered lunar scale; "Reported" is the median working error. CI is the 95% binomial interval for one-SE coverage; 1u and 1.96u denote the two reported coverage columns. These results do not use the auxiliary noise simulations. The final column counts $A_L > 3u$, not a separately calibrated discovery probability.

Case	True A_L	Bias	SD	Reported	1u cov.	CI	1.96u cov.	$A_L > 3u$
a	0	−0.002	0.175	0.183	0.69	0.638–0.745	0.97	0
a	1	+0.006	0.181	0.185	0.66	0.600–0.710	0.95	297
b	0	−0.012	0.175	0.181	0.71	0.652–0.758	0.95	0
b	1	+0.011	0.177	0.181	0.70	0.641–0.748	0.96	297
c	0	+0.006	0.180	0.185	0.72	0.662–0.767	0.95	1
c	1	+0.020	0.182	0.190	0.67	0.610–0.720	0.94	296
d	0	+0.002	0.175	0.187	0.67	0.617–0.726	0.94	1
d	1	+0.010	0.178	0.185	0.70	0.645–0.751	0.96	299

Extended Data Table 4 | Ten-day block support in 2026 auxiliary reconstruction. Assignment uses the mean bin time; boundary-crossing source members remain intact. The start is inclusive and the end exclusive. All dates are BJT in 2026. Block duration describes the model, not continuous uptime.

Block	Calendar start–end	Bins	Source members	Observation period
2046	01-07–01-17	403	241,474	January
2055	04-07–04-17	181	108,008	April
2056	04-17–04-27	259	153,774	April
2058	05-07–05-17	121	72,085	May
2059	05-17–05-27	299	178,683	May
2063	06-26–07-06	527	315,603	June–July
2064	07-06–07-16	72	42,708	June–July
2067	08-05–08-15	592	355,036	August a
2068	08-15–08-25	341	203,702	August b
2069	08-25–09-04	110	66,000	August b

Extended Data Table 5 | Auxiliary conditional simulation bias and standard errors. T: ten-day whole template without daily terms; L: M2+N2 with daily terms. Each Gaussian scenario uses 1,024 simulations; each Student/MA scenario uses 4,096 with fixed unequal period scales. Entries are bias and mean scale standard errors. These simulated responses do not certify arbitrary noise or selection effects. Corresponding interval coverage is in Extended Data Table 6.

Target / noise	Bias	LS SE	Day SE	Period SE
T / iid	-0.000	0.096	0.096	0.091
T / OU 1h	-0.007	0.094	0.264	0.255
T / OU 12h	-0.002	0.084	0.247	0.226
T / OU 3d	+0.003	0.064	0.146	0.134
T / period constants	+0.001	0.043	0.043	0.041
T / t7 MA	+0.002	0.088	0.074	0.071
T / t5 MA	+0.001	0.088	0.074	0.071
T / t7 MA + OU	+0.001	0.087	0.098	0.090
L / iid	-0.001	0.255	0.298	0.270
L / OU 1h	-0.002	0.238	0.799	0.714
L / OU 12h	+0.031	0.207	0.837	0.631
L / OU 3d	+0.004	0.164	0.594	0.359
L / period constants	+0.004	0.114	0.134	0.122
L / t7 MA	+0.004	0.236	0.244	0.213
L / t5 MA	+0.003	0.236	0.243	0.213
L / t7 MA + OU	+0.005	0.231	0.336	0.262

Extended Data Table 6 | Auxiliary conditional interval coverage. Nominal 95% coverage (%) for the same targets and simulations as Extended Data Table 5. Columns report individual LS/day/period t intervals, the $1.96u_{\max}$ interval, and wild true-value acceptance. Gaussian and Student/MA scenarios have 1,024 and 4,096 simulations, respectively. True-value acceptance is distinct from numerically inverting a test, and none of these conditional scenarios provides a universal coverage guarantee.

Target / noise	LS	Day	Period	Max-SE	Wild
T / iid	95.0	94.8	94.2	96.7	96.5
T / OU 1h	49.9	94.1	93.7	94.5	96.5
T / OU 12h	49.7	95.6	94.1	96.1	97.4
T / OU 3d	60.9	96.4	94.0	95.9	96.9
T / period constants	94.7	93.9	93.8	96.0	96.0
T / t7 MA	97.9	94.6	94.6	98.3	96.5
T / t5 MA	98.4	95.2	94.7	98.9	96.7
T / t7 MA + OU	93.7	96.4	94.9	97.2	97.0
L / iid	95.2	96.8	94.4	98.3	96.9
L / OU 1h	50.2	96.3	93.1	96.9	96.7
L / OU 12h	49.9	99.0	92.9	98.8	96.5
L / OU 3d	67.0	99.8	93.5	99.8	95.8
L / period constants	96.2	97.5	94.1	98.3	97.5
L / t7 MA	96.9	96.7	92.7	98.2	96.6
L / t5 MA	97.1	97.0	92.4	98.3	96.7
L / t7 MA + OU	93.1	98.5	93.0	98.5	96.9

Supplementary Methods

Sections S1–S4 specify the auxiliary 2026 reconstruction; S5 specifies the separate campaign analysis. They do not redefine the primary period-offset estimator or its reported uncertainty. No spliced or fitted-offset long-term stability result is included.

S1. Auxiliary ratio coordinate and membership

This auxiliary route uses 2026; the separately recorded 2025 conversion is retained for commissioning-stage diagnostics.[11] Let $N_{1397} = 858456$, $N_{1550} = 773598$, $N_{1156} = 1295739$ and $N_{1550,\mathrm{WH}} = 966996$. The 2025 chain gives

$$f_{1397} = \frac{f_{\mathrm{Sr}}}{2(1-\alpha)}, \quad \alpha = \frac{10.93629485}{25N_{1397}}, \quad f_{1550,\mathrm{WH}} = \left(N_{1550} + \frac{8}{25}\right)\frac{f_{1397}}{N_{1397}}.$$

The eight Shanghai reference units include seven in the transmitted carrier and one added at Wuhan. The Wuhan repetition frequency is $f_{\mathrm{Yb}}/(2N_{1156})$, its reference is $f_{\mathrm{Yb}}/(40N_{1156})$, and its comb offset is 26 reference units. For recorded frequency q in nominal hertz realized by that Wuhan reference,

$$R_{\mathrm{lock}} = \frac{N_{1156}}{N_{1397}}\frac{N_{1550} + 8/25}{N_{1550,\mathrm{WH}} + 26/20 + dq/(20\times 10^7)}\frac{1}{1-\alpha}.$$

We use $d = 2$ and the stated α in 2025, versus $d = 1$, $\alpha = 0$ in 2026. The respective counter values are near 66 and 33.6 MHz; the former represents a 132-MHz beat before division. The counter-reference convention is an experimental assumption, not implied by a 10-MHz label. The electronic divider is restored once, independently of any round-trip correction.

For signed Sr and Yb shifts a and b, $R = R_{\mathrm{lock}}(1+a)/(1+b)(1+\Delta G)$, with inherited $\Delta G = -3.116\times 10^{-15}$. We form $y = (R/R_{\mathrm{ref}} - 1)10^{18}$ using $R_{\mathrm{ref}} = 1.2075070393433377166$ and Decimal anchors, avoiding subtraction of full ratios in float64. Positive y means faster Wuhan Yb relative to Shanghai Sr. For positive Newtonian potential U, the rate contribution is $-[U_{\mathrm{WUH}} - U_{\mathrm{SHA}}]/c^2$. These conventions do not establish a new absolute ratio or static potential.

The eight periods are September/October 2025, January/April/May 2026, June–July, and early/late August. In 10^{-18} units, Sr density corrections are [−0.43,−0.41,−0.30,−0.33,−0.25,−0.24,−0.33,−0.29], the first six Zeeman corrections are [−178.50,−178.87,−179.14,−180.19,−180.18,−178.57], and the background-gas term is −0.67. August Zeeman and all blackbody corrections are recorded as online. The first five AC entries are missing; the final three are [7.44,8.93,9.03]. Positive 37.2-mHz feedback applies to the first three periods at the assumed 698-nm optical node, giving 86.6119×10^{-18}—larger than the approximately 10×10^{-18} tide. Printed Yb shifts for original groups 1–9 are primary, with component sums tested separately; May Yb is missing. Computational zeros for unknown entries are not measured zeros. Missing terms have no certified bound. The ledger is frozen for these refits, but neither its precedence over all earlier residual inspection nor within-block constancy is established. Baselines absorb constants; time-dependent shifts can project onto the tide.

Commissioning records and corrections Reported commissioning, processing changes and unresolved offsets motivate separate September and October 2025 stage- or interval-intercept analyses, without shared amplitudes across months or with 2026. This scope was clarified during reanalysis.

An older uncompensated Yb shift near -1.1×10^{-17} may overlap the updated totals -9.7 and -9.5×10^{-18}, which include a lattice entry of -8×10^{-18}. Inclusion is unverified; we do not add it twice.

An additional δb multiplies R by $(1+b)/(1+b+\delta b)$, so $\delta b = -1.1 \times 10^{-17}$ would increase R by approximately that fractional magnitude. Additive-shift invariance of intercept designs does not verify this multiplicative correction or its provenance. September and October stage fits give 0.87 ± 0.67 and 1.57 ± 0.26 (LS only), with larger errors after daily controls. Only two days per month preclude establishing correlated-noise coverage; these are commissioning diagnostics.

Observation selection and averaging The corrected intervals select 1,966,912 unique source rows from 42 files, retaining fractional-second BJT labels. Windows include the first PDF second and exclude the second following the last. Rows are time/file/row sorted and deduplicated by source identity. Finite offsets below 1 MHz in magnitude and strictly within 10 Hz of the plausible-sample median leave 1,966,908 rows. The three-row difference from the experimental count retains boundary provenance.

Within each interval, 300-s blocks start at the first source timestamp surviving the initial screen. Their means are compared with a 13-block running median with reflected boundaries. Empty blocks retain their fixed-grid positions as NaN means, ignored within each median window. Residual median absolute deviation divided by 0.6744897501960817 sets the scale. Blocks exceeding six scales and their neighbours are excluded, removing 11,131 rows without replacing retained values or using tidal residuals. Averaging instead starts at the floored earliest source timestamp before screening: half-open 600-s windows never cross effective intervals. Every window, including a partial tail, requires at least 300 retained members. The 3,266 means contain 1,952,992 members. Restricting to 2026 leaves 2,905 means and 1,737,073 members; September/October 2025 retain 160/201 means. Theory columns use identical members. Baseline indicators use mean bin times, retaining complete boundary-crossing bins; observations are not interpolated. Day deletions use mean-bin BJT days, not every raw sample of that civil day: the parent record includes 16 midnight-crossing bins. Retained 2026 samples occupy every local hour, with 45.6% between 06:00 and 18:00. Extended Data Table 4 lists ten-day support. No block crosses the year-convention change, and within-block drift tests identify no specific hardware event.

We repeated the fixed designs using no local screening and the retained six- and eight-scale rules. These leave 2,916, 2,905 and 2,909 bins, respectively, retaining the initial validity rules and minimum membership. Each scenario averages response and theory over identical members. Scalar and joint two-coefficient tests are separate; masks also change bin-mean times and design. The auxiliary six-scale rule is unchanged. These correlated sensitivities neither identify physical lock failures nor calibrate screening-selection uncertainty.

S2. Auxiliary prediction and physical sensitivities

The common prediction combines solid-Earth and ocean-loading series on a 90-s grid. Pysolid displacements under the recorded IERS 2010 implementation[18] are converted using

$$s(t) = -\frac{1+k_2-h_2}{h_2}\frac{g[u_{\mathrm{WUH}}(t)-u_{\mathrm{SHA}}(t)]}{c^2},$$

with $h_2 = 0.6078$, $k_2 = 0.3010$ and $g = 9.794\,\mathrm{m\,s^{-2}}$. This displacement approximation is not a complete degree-dependent potential calculation. Loading uses 18 constituents from grids locally documented as FES2022b.[19] The FES2014 paper describes an earlier atlas, not validation of FES2022.[20] FES2022 was funded by CNES, produced by LEGOS, NOVELTIS and CLS, and made freely available by AVISO. Convolution uses PREM load-Love Green functions containing $1+k'_n-h'_n$; the Han–Wahr table[22] through degree 696 is extended to 40000, tapering the final 30%, with the recorded centre-of-solid-Earth degree-one convention. Direct water potential, induced potential and displacement are included. Member-matched solid, loading and total demeaned RMS values are $(3.245, 1.507, 4.410) \times 10^{-18}$. Loading/total

RMS is 34.2%, with solid–loading correlation 0.680; variances are not additive. Atmospheric loading is neither modelled nor bounded. Free daily terms control empirical variation but do not validate an atmosphere-loading field, nor can local pressure differences do so.

The 360,001 theory samples span 1 September 2025, 08:00 to 11 September 2026, 08:00 BJT, inclusive. Equal-weight least squares over this full grid projects onto quadratures of Q1, O1, P1, K1, N2, M2, S2, K2, Mf, Mm, Msf, Sa and Ssa plus a constant (rank 27), retaining the remainder. BJT converts to UTC once, with the saved FES convention and TT−UTC=69.184 s. Harmonic columns, including the lunar quadrature, are evaluated on the 90-s grid and interpolated before source-member averaging. Whole-template fits scale the complete prediction; component fits subtract the remainder once. In 10^{-18} units,

$$y - F = A_L C_L + a_{12} C_{12} + a_D C_D + Z_b \gamma + Z_{24,12} \beta + \epsilon.$$

C_L, C_{12} and C_D contain M2+N2, S2+K2 and the four diurnal constituents; F contains remaining theory, including long-period terms. The daily design has sine/cosine columns at one and two cycles per day in each occupied block. The ten-day lunar design has 53 parameters and full rank. Blocks follow $\lfloor t/(10 \times 86400) \rfloor$ on the BJT epoch coordinate; five-day blocks follow the analogous rule. S2+K2 is not interpreted as resolved from exact 12-h terms. Drift tests add a within-block linear column, centred on observed mean time, on identical data; paired deletions estimate changes. These short-period baselines/slopes never enter seasonal fits.

Twenty-day, thirty-day and per-period controls give lunar scales 0.867 ± 0.246, 0.796 ± 0.202 and 0.761 ± 0.205 from the same sample time. Thirty-day blocks share a baseline/daily pattern across August periods; per-period models separate them. Their scalar wild $p = 0.03125$ does not motivate replacing the weaker five-/ten-day auxiliary controls (joint tests: the auxiliary phase analysis in Methods).

Three sensitivities replace only the solid-Earth series with DE430 geometric potentials:[28] degree two with response $1 + k_2 - h_2$; degrees two/three with constant responses; and degrees two/three with nominal order-dependent real k_{nm} from IERS Table 6.3.[18] We use $h_3 = 0.292$, IAU 2006/2000A rotation, UT1≈UTC and zero polar motion. These first three alternatives hold the response coefficients constant in frequency. Whole-template scales become 0.876, 0.872 and 0.872; lunar scales become 0.895, 0.887 and 0.887. For the degree-two constant-response alternative, substituting Hou et al. coordinates[11] (Wuhan 30°31′ N, 114°19′ E, 30 m; Shanghai 31°14′ N, 121°29′ E, 5 m), with the same potential algorithm and loading, changes A_T by +0.0066 and A_L by +0.0046; solid-template difference RMS is 0.043×10^{-18}. Neither coordinate set is certified at the clock transition points; height references also require confirmation.

A separate finite-basis sensitivity applies nominal frequency-dependent $\gamma(\omega) = 1 + k(\omega) - h(\omega)$ from the IERS resonance and anelastic approximations[18] implemented in pyTMD 3.0.9. We project the constant-response degree-two DE430 series onto the same 13 constituents, transform each pair by γ/γ_0, and retain the constant and residual from each projection and the original loading series. Here $\gamma_0 = 1 + 0.3010 - 0.6078$. Both projections use equal weights; transformed series are evaluated on the original full grid and averaged over the unchanged source members. For $C \cos\theta + S \sin\theta$ and $\gamma/\gamma_0 = a + ib$, the transformed coefficients are $(aC + bS, aS - bC)$. Relative to this same DE430 constant control, real-only and complex responses change the baseline-only whole-template scale by -0.00442 and -0.00434, respectively; changes in daily-controlled A_L are below 0.00008. Trimming 30 days from each end of the theoretical projection window changes these scales by less than 0.000007. The observations and reported primary estimates are unchanged; alternative estimates are retained as sensitivities. This nominal finite-basis check does not reproduce the full effective IERS response, separate nodal-sideband responses, latitude dependence or complete ocean feedback; its small shifts do not bound all theory errors.

S3. Auxiliary covariance, resampling and noise simulations

Unweighted OLS uses column-normalized SVD. With n averages, $p = \mathrm{rank}(X)$, residuals r and the original-scale pseudoinverse $B = X^{+}$, $C_{\mathrm{LS}} = [\sum_i r_i^2/(n-p)]BB^T$. Delete-day and delete-period covariances are also retained. Long-period analyses additionally retain elapsed-time Bartlett HAC sensitivities at one, three and ten days. Jackknife covariance[23] is $C_{\mathrm{JK}} = (G-1)G^{-1}\sum_g(\hat{\theta}_{-g}-\bar{\theta})(\hat{\theta}_{-g}-\bar{\theta})^T$. Deletion-group independence remains an assumption. Unsupported nuisance columns are removed; target/rank loss is recorded. The maximum LS/day/period SE is a scalar summary, not a joint covariance. Two-coefficient regions use $(b-\hat{b})^T C^{-1}(b-\hat{b}) \le 2F_{2,\nu}(0.95)$, with $\nu = G-1$ for jackknife covariance and $n-p$ for LS. Extended Data Fig. 2b uses delete-day covariance with 34 denominator degrees of freedom.

Restricted wild tests[27] fit each null and enumerate 64 period-wise sign patterns, studentizing with delete-period covariance. Scalar tests use bootstrap-t; joint phase tests use both coefficients and their covariance (the auxiliary phase analysis in Methods). Period independence and error symmetry are working assumptions; nuisance estimation prevents exact randomization inference. The minimum two-sided p-value is 0.03125. Exploratory choices are not multiplicity-adjusted. Scalar inversion partitions the real line at polynomial roots, including endpoints and isolated-point checks. The ten-day daily-term whole-template 95% set is $[-0.541,-0.294]\cup[0.241,0.367]\cup[0.385,2.568]$, not its convex hull. Independent QR and covariance checks verify implementation agreement, not experimental coverage.

Fixed-design simulations retain actual times: 1,024 per design/scenario, seed 202611052. Errors are Gaussian iid variance 500 in working units; white SD 10 plus OU SD 20 with timescale 1 h, 12 h or 3 days; or white SD 10 plus independent period constants SD 30. Constants may be absorbed by nuisance columns and do not constitute a general drift test. Extended Data Tables 5 and 6 report ten-day whole-template and lunar performance. $\hat{A}\pm u_{\max}$ is a one-SE summary; normal and t_5 multiples define separate nominal 95% intervals. True-value wild acceptance is distinct from numerical inversion. The historical 124/200 injection coverage concerned one SE, not 95%, and cannot certify extreme-tail significance.

A Gaussian white-plus-OU diagnostic averages exponential covariance over actual member pairs, with white-bin variance proportional to $600/n_i$. ML/REML profile a common scale over iid and 70 combinations of doubling timescales 10–5120 min and mixing fractions 0.1, 0.3, 0.5, 0.7, 0.9, 0.97 and 0.995. Whitened residuals show lag-one correlations −0.13 to −0.17, period RMS ratios 1.37–1.38 and kurtosis 4.82–5.12. None of 4,096 conditional simulations per selected model exceeds the observed maximum absolute lag-1–12 correlation, RMS ratio or kurtosis. They retain mean projection but do not reselect covariance parameters; these are not selection-adjusted probabilities. This noise family is inadequate for those diagnostics, and its likelihood significance is not used for detection.

Three further stress scenarios use variance-normalized Student-t7 or t5 innovations, negative MA(1) correlation −0.15, and fixed period scales from whole-template residuals. A third mixes t7 MA and member-averaged 12-h OU at variance fractions 0.9/0.1, adjusting MA correlation to approximately preserve lag one. MA memory restarts across interval changes or gaps ≥900 s. White-bin variance scales as $600/n_i$; period scales are $[18.518, 17.888, 19.313, 19.170, 24.699, 21.638]\times 10^{-18}$. Each has 4,096 simulations (seed 202611056), without re-estimating these empirical parameters or reselecting samples. Delete-day joint-phase regions cover 96.8–98.2%, versus 85.8–86.9% for delete-period ellipses and 96.8–96.9% wild acceptance. Joint wild power for unit response is 19.9–27.6%, versus 99.0–99.8% for the whole template without daily terms. These exploratory conditional tests neither identify instrument noise nor certify arbitrary systematics or model selection.

Additional short-period diagnostics The whole-template point estimate is 0.8836; LS, delete-day and delete-period errors are 0.0882, 0.1002 and 0.0971. For the lunar scale 0.8909, they are 0.2344,

0.2824 and 0.1881. The ten-day block-drift change in the whole-template scale is −0.006 ± 0.030 under paired day deletions; five-day blocks with drift give 0.868 ± 0.112. The ten-day daily-term whole-template zero-response p-value is 0.03125, versus 0.0625 with five-day blocks. Fixed-phase lunar p-values are 0.0625 and 0.09375, respectively. All are conditional, two-sided period tests; only 64 sign patterns are available.

Across no local screening and the six-/eight-scale screens with five-/ten-day baselines, whole-template scales without daily terms range from 0.884 to 0.898 and lunar scales from 0.887 to 0.991. Removing local screening changes the ten-day daily-term whole-template p-value from 0.03125 to 0.0625. These are sensitivities on overlapping observations, not alternative independent detections.

The six-period auxiliary held-out comparisons share training observations.

S4. Level-preserving long-period bounds

Long-period models jointly fit five quadrature pairs, short-period group scales and global daily terms on 2026 data with one intercept. Alternatives add one global drift, use component-sum corrections or omit May. No segment, period or month intercepts are used; all 2025 measurements are excluded.

The no-drift individual 95% amplitude bounds for Mf, Mm, Msf, Sa and Ssa are

$$(1.60, 4.32, 2.70, 20.98, 11.11) \times 10^{-17}$$

(Extended Data Fig. 3b). A global drift expands Sa and Ssa bounds to approximately 6.0×10^{-15} and 1.1×10^{-15}. The no-drift Sa bound is about 75,000 times the current predicted amplitude. Neither the retained-level plotting coordinate nor these bounds represents the calibrated absolute-ratio coordinate of the earlier summer analysis.

For a pair $\hat{b}$ with covariance C and denominator degrees of freedom ν, a conservative radial bound from the nominal ellipse is

$$U = \|\hat{b}\| + \sqrt{2F_{2,\nu}(1-\alpha)\lambda_{\max}(C)}.$$

We use $\alpha = 0.05$ for an individual bound and 0.01 for the five-target Bonferroni family. With six period clusters, $\nu = 5$ is a working approximation. The no-drift family upper bounds for Mf, Mm, Msf, Sa and Ssa are $(2.30, 6.18, 3.75, 30.38, 15.28) \times 10^{-17}$. Omitting May leaves five periods and gives individual Sa and Ssa bounds of approximately 1.37×10^{-15} and 9.36×10^{-16}. Bounds are conditional on the calibration coordinate; missing early Sr AC corrections and other unbounded changes prevent an unconditional gravitational limit.

The independent solar check averages the degree-two potential over hour angle. With P_2 the second Legendre polynomial, it evaluates $-(1+k_2-h_2)GM_\odot[r_W^2 P_2(\sin\varphi_W) - r_S^2 P_2(\sin\varphi_S)]P_2(\sin\delta)/(r_\odot^3 c^2)$ using DE430[28] and the archived theory-input coordinates (Wuhan: 30.529° N, 114.357° E, 50 m; Shanghai: 31.130° N, 121.570° E, 20 m), converted to geocentric radii and latitudes for this calculation. This checks the small Sa/Ssa scale without fitting observations. It does not provide a complete seasonal-loading model. The existing 18-constituent loading series contains none of Mf, Mm, Msf, Sa or Ssa.

S5. Separate campaign analysis and conditional injections

The author-supplied campaign analysis is a separate 54-segment route with about 544 retained hours. Its selection first uses a 10-Hz screen, retains the longest uninterrupted run, and trims terminal ex-

cursions exceeding 1% of the peak-to-peak range from the median. A local 300-s block screen uses a ±30-min running median and six robust standard deviations, also excluding neighbouring blocks. This removes about 0.4% (2.2 h); the primary P8 mask does not use this screen. Adding −1, +1 or +3 times the tide before screening leaves the selected mask unchanged at the selected threshold. This is an injection check, not proof that the study was prospectively blind. Without this local screen the campaign lunar estimate is 0.77±0.19; omitting only end trimming changes the lunar scale by −0.005 and whole-template scale by +0.008.

The campaign prediction uses a physical potential construction with degree/order-dependent solid-Earth response and eight FES2022 loading constituents. Its M2 solid/loading/total amplitudes are 3.41/1.42/4.77 in 10^{-18} units, distinct from the primary synthesis. The source loading integration uses the PREM-based response, fine local coastal resolution and coarse far-field blocks; S2 loading differences remain a recorded sensitivity. These source-level physical and simulation claims have not been independently rerun for this synthesis.

The whole-template model is $\Delta y_j = c_{i(j)} + A_T T_j + \epsilon_j$. For the lunar fit, M2+N2, S2+K2 and the diurnal group receive independent scales, and each campaign receives independent sine/cosine terms at exactly 24 h and 12 h. The eight-constituent decomposition explains 99.84% of model variance; its residual, RMS 0.18×10^{-18}, is retained at its model value. The reported uncertainty for solar-day-controlled fits is the larger of residual-scaled least-squares and delete-day jackknife errors.[23] The two errors and the residual-resampling spread are distinct quantities.

Residual resampling circularly shifts fitted residuals within each continuous filtered run and adds them to a unit-response model. Two hundred realizations give an SD of 0.16, bias −0.01 and one-SE coverage of 0.69 for the campaign lunar estimator. The no-daily-term whole-template, semidiurnal and lunar errors in the source campaign figure are resampling SDs. Resampling 29 operating groups, or 54 segments where specified, with replacement uses 1,000 realizations and assumes independent resampling groups.

Solar-day injection scenarios use the fitted campaign terms with (a) fitted amplitudes/phases, (b) 1.5-times amplitudes and random phases, (c) daily phase jitter of 45° RMS, or (d) phase random-walk steps of 20° per day and amplitude steps of 10% per day. Each scenario has 300 realizations at true lunar scale zero and one, seed 20261116. Extended Data Table 3 retains the bias, SD, reported error and coverage for every case. A further 20,000 zero-response realizations, 5,000 per scenario, estimate the upper-tail exceedance of the observed point estimate: zero exceedances gives the one-sided 95% binomial upper bound $1-0.05^{1/20000} \simeq 1.5\times10^{-4}$. It is not a two-sided cluster-test p-value or a probability that the null is true. Correlations within runs are preserved by this resampling; more general inter-run dependence and arbitrary systematic errors are not thereby calibrated.

Splitting summer solar-day terms by month, adding their linear drift and deleting campaigns give the sensitivities in Extended Data Fig. 1. Campaign full-template leave-period-out comparisons give improvement in 7/8 periods without daily terms and 5/8 with them, with pooled residual-square reductions of 4.9% and 0.6%; held-out offsets are fitted locally. These are separate from the six-period auxiliary results and are not combined. An eight-period wild-test value in the source draft is not used for inference here pending verification of its statistic, sign enumeration and two-sided tie counting; it is not replaced by an auxiliary p-value.

S6. Interpretation across routes

The primary P8 estimate, campaign daily-controlled estimate and 2026 auxiliary estimate use different response vectors, selection, averaging, template conventions and nuisance designs. Their central values are compared without pooling them. No p-value, coverage rate, noise scale or calibration status is

transferred between routes. The P8 phase-diversity claim is supported by its own linear-estimator gains, conditional on the stated disturbance class. The campaign injections and auxiliary OU/Student/MA simulations characterize their own estimators only. Conditional performance does not establish control of arbitrary environmental disturbances.

All long-term stability curves derived from fitted offsets or spliced residuals are excluded. The global-intercept long-period analysis in S4 is an amplitude-bound calculation on retained levels, not a stability analysis. Accumulated exposure and cumulative amplitude estimates likewise do not establish uninterrupted stability.